

TANKS CALCULATED AVERAGE

Product	NAPHTHA
Port	BILBAO, SPAIN
Vessel	TORM SUBLIME
Laboratory	PETRONOR/INTERTEK
Nº of Tanks	4
Observations	Volumetric calculated quality MEETS contractual parameters.

Test	Method	Units	Average
DENSITY 15º C	ASTM D 4052	Kg/L	0,6838
DISTILLATION :	ASTM D 86		
INITIAL BOILING POINT	ASTM D 86	ºC	31,2
05% RECOVERED	ASTM D 86	ºC	43,9
10% RECOVERED	ASTM D 86	ºC	47,0
50% RECOVERED	ASTM D 86	ºC	67,4
90% RECOVERED	ASTM D 86	ºC	106,6
95% RECOVERED	ASTM D 86	ºC	123,0
END POINT	ASTM D 86	ºC	141,0
RESIDUE	ASTM D 86	%vol	1,0
SAYBOLT COLOR	ASTM D 6045	-	30,0
REID VAPOR PRESSURE	ASTM D 5191	psi	10,9
SULPHUR	ASTM D 5453	mg/kg	18,9
MERCAPTAN SULPHUR	ASTM D 3227	ppm	5,4
CLORINE	UOP 779	ppm	0,6
LEAD	IP 224	ppb	<20
MERCURY	UOP 938	ppb	<0,5
HIDROGEN SULFIDE	UOP 163	ppm	1,06
PARAFFINS	EN 14517	%vol	80
OLEFINS	EN 22854	%vol	0,4
NAPHTENES	EN 22854	%vol	12,7
AROMATICS	EN 22854	%vol	7,2
MTBE	CG	ppm	1,1
TOTAL OXIGENATES	ASTM D 7423	ppm	3,7
BENCENE	EN 22854	%vol	4,2
ISO - PARAFFINS	EN 22854	%vol	42,1
NORMAL - PARAFFINS	EN 22854	%vol	38

This report is only for information and guidance. Calculations have been intended solely for the estimation of the volumetric quality of the referenced cargo as required by the client. Volumetric sample has not been prepared nor analyzed. The volumetric averages are calculated from the proportional amounts loaded from the individual tanks:

Tank 406 (5,5%) Tank 405 (41,6%) Tank 408 (38,6%) Tank 408 BIS (14,3%)

Please consider that:

- Tanks results have been tested and provided by PETRONOR/INTERTEK Laboratory Staff
- According to IFIA bull 06-01 results must be issued only on PETRONOR/INTERTEK paperwork, for more information please refer to laboratory report. Intertek bears no responsibility for accuracy of the results calculated from analysis.